Aditya Kashyap

+91-8368935139 | aditya24kashyap@gmail.com | LinkedIn | GitHub | LeetCode

Summary

Final-year B.Tech (ECE) student and full-stack engineer shipping production, end-to-end apps spanning UI, backend, database, and LLM/AI integration. Built and deployed four full-stack products (Next.js/TypeScript/PostgreSQL/MongoDB) integrating LLMs for retrieval, code generation, and AI-assisted workflows, including a RAG pipeline and vector database built from scratch. Strong systems/infra background in C++, multithreading, and networking (TCP/IP, TLS). Solved 300+ LeetCode problems (rating 1800+, top 10%); active open-source contributor.

Technical Skills

Languages: C++ (primary), JavaScript, TypeScript, Python, SQL
AI / LLM: LLM Integration (Ollama, Google Gemini API), RAG Pipelines, Prompt/Streaming Completions, Embeddings (nomic-embed-text), Vector Search (HNSW, KD-Tree)
Full-Stack / Frontend / Backend: React.js, Next.js 15 (App Router, Server Actions), Node.js, Express.js, REST APIs, WebSockets, Prisma ORM, Tailwind CSS, shadcn/ui, TanStack Query, Zustand
Databases: PostgreSQL, MongoDB, MySQL
Systems / Infra: OS, Computer Networks, Multithreading, TCP/IP, TLS, Producer-Consumer Architectures, Docker (basics)
Auth & DevOps: OAuth 2.0 (NextAuth, Better Auth), Git, GitHub, Vercel, Linux, Postman

Experience

Open Source Contributor | *GirlScript Summer of Code (GSSoC Extended 2025)* *Jul 2025 – Oct 2025*

– Merged pull requests across open-source repositories fixing bugs, adding features, and improving documentation; collaborated with maintainers through code reviews in a program with 10,000+ contributors, demonstrating high ownership and independent execution.

AI/ML Training Program | *YBI Foundation* *Sep 2025 – Oct 2025*

– Built end-to-end ML pipelines for sentiment analysis and predictive analytics using Python, Scikit-learn, Pandas, and NumPy; applied classification, feature engineering, and model evaluation on real-world datasets.

Projects

Browser IDE *Next.js 15, TypeScript, WebContainers, MongoDB, NextAuth, Ollama*

– Built a full-stack, production-deployed browser IDE (UI + backend + DB + LLM) with Monaco Editor, in-browser Node.js execution via WebContainers API, and an xterm.js terminal so users write, run, and debug code with zero local install.
– Architected a modular AI provider layer integrating local LLMs (Ollama/codellama) with streaming completions, swappable to Google Gemini for cloud deployment with a single config change — core code-intelligence / AI infra work.
– Shipped OAuth 2.0 auth (Google & GitHub via NextAuth v5), MongoDB session persistence via Prisma, Zustand global state, and 6 bootstrapped project templates (React, Next.js, Vue, Express, Hono, Angular).

PostAPI *Next.js 15, TypeScript, Prisma, PostgreSQL, TanStack Query, Zustand, Gemini*

– Built and deployed an end-to-end Postman alternative (UI + backend + PostgreSQL + AI) with a REST client (methods, headers, params, body, response viewer) and a WebSocket client for real-time `ws:///wss://` testing.
– Implemented multi-workspace collaboration with invite links and role-based access (Admin/Member), OAuth 2.0 via Better Auth (GitHub & Google), and a Monaco-powered raw request/body editor with JSON validation.
– Integrated the Google Gemini API to power AI-assisted request building and response analysis; used Prisma ORM over PostgreSQL and TanStack Query + Zustand for server-state caching and persistent client state.

CortexDB *C++17, Python, HNSW, REST API, Ollama, RAG*

– Implemented a vector database from scratch with HNSW (O(log N), the algorithm used by Pinecone and Weaviate), KD-Tree, and brute-force search across cosine, Euclidean, and Manhattan distance metrics.
– Built a full RAG pipeline: 768D embeddings via nomic-embed-text, overlapping document chunking, HNSW semantic retrieval, and LLM-generated answers via llama3.2 — all served over a REST API, showcasing AI infra fundamentals.

Packet Inspector *C++17, POSIX Threads, TCP/IP, TLS*

– Built a multithreaded Deep Packet Inspection engine that parses PCAP files, extracts TLS SNI from Client Hello packets, and classifies HTTPS traffic (YouTube, Facebook, etc.) without decryption using a producer-consumer thread pool with consistent hashing and mutex/condition-variable-protected queues for zero-race-condition state tracking.

Education

Maharaja Surajmal Institute of Technology *Delhi, India*

– B.Tech in Electronics and Communication Engineering — CGPA: 8.95 2022 – 2026

Mahashay Chuni Lal Saraswati Bal Mandir Sr. Sec. School *Delhi, India*

– Class XII (CBSE) — 78.8% (2021) | Class X (CBSE) — 86.6% (2019)